



SYMBIOSIS INSTITUTE OF TECHNOLOGY

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Course Name: DevOps Lab
Course Code: TE7950
Faculty: Engineering
Course Credit: 2
Course Level: 4
Sub-Committee (Specialization): Computer Science

Learning Objectives:

1. Understand the working of projects in the devops pipeline. Will understand the application deployment system using docker containers.
2. Understand Version Control, Branching development model using GIT. Build and maintain the Jenkin Pipeline.
3. Working with Docker Image with DTR, with multiple web applications.
4. Will understand the Infra challenges and using Kubernetes. Will be able to create mongo and mongo express deployment.
5. Will understand the concept of microservices and linking with pipelines.

Pre-requisites:

Python / Powershell / Perl / Java / .net, scripting, cloud technology, software engineering

Course Outline:

Unit No.	Details	Hours
1	Project 1: Dockerizing Jenkins Pipeline- Demonstrate continuous integration and delivery by Dockerizing Jenkins Pipeline.	6
2	Project 2: Deploy Angular Application in Docker Container-Deploy an Angular application in Docker. The Angular application should be built with the Angular CLI along with Docker Compose for development and production.	6
3	Project 3: Branching Development Model-Build a branching model to help your team understand the git workflow for faster work integration.	6
4	Project 4: Architecting Jenkins Pipeline for Scale-Use Jenkins to set	6



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	up a distributed pipeline that will compile and test a Maven project on two different slave nodes respectively.	
5	Project 5: Containerizing application and scanning its Docker image with DTR-Deploy a spring boot application on Docker for a retail company with multiple web applications.	6
6	Project 6: Social Media underlying infra challenges-Using Kubernetes, complete the features of application scalability by demonstrating how to create a cluster with the autoscale feature.	6
7	Project 7: Create Mongo and Mongo Express deployments ,services,configmaps and secret	6
8	Project 8: Create deployments ,services,configmaps and secrets to containerize a complete application with at least 4 microservice	6
9	Project 9: Create apache2 server within a deployment and access it using host machine using commands learn using K8s	12

Recommended Books:

Sr. No.	Book title	Authors	Publishers
1	A Practical Guide to Continuous Delivery	Eberhard Wolf	Addison-Wesley
2	Devops with windows server 2016	Ritesh Modi	PACKT
3	Implementing Devops with Ansible 2	Joathan McAllister	PACKT
4	The Devops 2.0 Toolkit	Viktor Farcic	PACKT
5	A Practical Guide to Continuous Delivery	Eberhard Wolf	Addison-Wesley

Pedagogy:

1. In LAB as HANDS ON session
2. Hands on Exercise
3. Capstone Projects



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Evaluation:

A. Continuous Assessment

Lab Assignment-Journal Write up

Viva

B. End Semester Examination

Project Presentation

Viva

Benchmarked against similar courses in other national/ international universities /organizations:

Sr. No.	Name of the Course	Name of University where it is offered
1.	Advanced Certificate Program in Devops	IIIT Bangalore
2.	PG program in Devops	Belhaven University

Name of the experts designing the course

Sr. No.	Name	Designation	Organization/ Institute
1	Dr. Kailash Shaw	Associate Professor	SIT
2	Mr. Yash Shah	Technology Mentor	SWABHAV Tech Lab